LALITH KUMAR J

+91 9360068491 | lalith.j2004@gmail.com | linkedin.com/in/lalithkumar52 | Salem, Tamil Nadu

SUMMARY

Electronics and Communication Engineering undergraduate with hands-on experience in embedded systems, IoT, Arduino, and sensor interfacing. Built and deployed four end-to-end embedded/IoT projects spanning deep learning, sensor automation, and assistive technology. Seeking an entry-level Embedded Systems Engineer role to apply strong fundamentals in C, sensor interfacing, and communication protocols.

TECHNICAL SKILLS

Programming: C, Core Java

Embedded Systems: Arduino, IoT, Sensor Interfacing, Microcontrollers

Protocols: UART, I2C, Wi-Fi, Bluetooth

Tools: Arduino IDE, VS Code, Code::Blocks

PROJECTS

Smart Sugarcane Disease Detector using Deep Learning — Apr 2026

- Developed a deep learning image-classification model to detect sugarcane leaf disease from smartphone photos, trained across 5 classes (Healthy, Rust, Yellow, Dot, and Mosaic).
- Trained the model on a labeled dataset and validated performance using accuracy/loss graphs, enabling instant disease identification from an input image for smart-farming decisions.

Smart Railway Sanitation System (IoT) — Mar 2025

- Designed an IoT-based smart toilet monitoring system for railway stations using sensor-based automation, reducing manual inspection rounds.
- Enabled real-time sanitation status alerts to improve hygiene response time and optimize maintenance scheduling.

Smart Blindness Assistance Glass (IoT, Embedded Systems) — Jan 2024

- Built an Arduino-based wearable assistance system using ultrasonic sensors to detect obstacles within a 4-feet range.
- Enabled real-time audio/haptic alerts for safer navigation for visually impaired users.

INTERNSHIPS

Embedded Systems Intern — DEV Technology Solutions | Jun 2024

- Worked on sensor interfacing, microcontroller programming, and IoT mini-projects as part of a team-based internship.

IoT Development Intern — Optimus Tech | May 2023

- Completed a two-week internship covering core IoT concepts and real-world application design.

EDUCATION

Bachelor of Electronics and Communication Engineering — 2022–2026
Knowledge Institute of Technology, Salem | CGPA: 7.02 / 10

Wisdom Matric Higher Secondary School, Salem
HSC: 83% (2022) | SSLC: 66% (2020)

ADDITIONAL INFORMATION

Certifications: Cloud Computing — NPTEL

Achievements: Circuit Debugging Competition (Sona College of Technology, Salem); Paper Presentation (K. Ramakrishnan Engineering College)

Languages: Tamil, English